RIT | Golisano College of Computing and Information Sciences
School of Information

NSSA322 Systems Administration II

Lab 3: Enterprise Storage & Automation

Student Lab Guide — TrueNAS SCALE 25.10 • ZFS • SMB / NFS • Snapshots & Replication • Cronicle

Group number	
Member 1 (A) name / RIT username	
Member 2 (B) name / RIT username	
Member 3 (C) name / RIT username	
Bookstack Lab 3 page URL	
TrueNAS FQDN / IP	
Cronicle FQDN / IP	

Instructions

This is a group lab. Complete the work on your RLES infrastructure during the Thursday 5:00–7:00 PM session. Document as you go in your shared Bookstack — not after you leave the lab. Partial credit is available for activities and documentation at the instructor's or TA's discretion.

All NSSA322 labs are scored out of 100. You must earn 70 or higher to pass this lab. Failing three labs (score below 70) results in automatic course failure, consistent with the NSSA-221 policy.

Do not do the following
Do not install TrueNAS, Cronicle, extra file services, or extra disks on the FreeIPA/IdM server, the Zabbix server, or the Graylog server. Do not share the ZFS pool root dataset. Do not disable SELinux on Rocky hosts. Do not use a TrueNAS 26.x BETA image unless the instructor posts that exception.

Introduction

Enterprise systems administrators are expected to protect data, publish it safely to mixed Windows and Linux clients, and automate the checks that used to live in ad-hoc cron jobs. This lab builds on the hybrid directory foundation from Lab 1 and the observability stack from Lab 2.

Your group will deploy TrueNAS SCALE as a dedicated storage appliance, design a redundant ZFS pool and purpose-built datasets, join the NAS to Active Directory, publish SMB and NFS shares, protect those datasets with periodic snapshots and local replication, and stand up Cronicle as a web-based scheduler for operational jobs.

You will make real design decisions: vdev layout, dataset purpose, snapshot retention, share permissions, and which work belongs in TrueNAS native tasks versus Cronicle. Those decisions belong in Bookstack, with evidence.

Lab Summary

Review Labs 1–2 → Provision TrueNAS SCALE and a Cronicle host → Create a redundant ZFS pool and child datasets → Join Active Directory and publish SMB + NFS → Configure snapshots and local replication → Recover a file from a snapshot → Deploy Cronicle and run one storage-validation job → Hook the NAS into Zabbix or Graylog → Document everything in Bookstack.

Goals

By the end of this lab, you will be able to:

- Deploy TrueNAS SCALE in RLES with a dedicated boot disk and separate virtual data disks.
- Design and create a redundant ZFS pool (mirror recommended) and child datasets that match intended share types.
- Join TrueNAS to the Lab 1 Active Directory domain and publish an SMB share with group-based ACLs.
- Export an NFS share and mount it from a Rocky Linux client, including an fstab/_netdev entry.
- Configure a periodic ZFS snapshot task and a local replication task, then recover a file from a snapshot.
- Deploy Cronicle on Rocky Linux and schedule at least one operational job that validates storage or share health.
- Add the NAS to the Lab 2 observability stack at a basic level (Zabbix host and/or Graylog search).
- Maintain professional collaborative documentation in your semester group Bookstack.

Prerequisites

- Lab 1 complete: hybrid IdM/AD trust, DNS/DHCP, and pfSense routing are working.
- Lab 2 complete: Zabbix and Graylog web UIs still load. Leave those VMs running.
- RLES capacity for one TrueNAS SCALE VM and one Rocky Linux 9 VM for Cronicle.
- Windows 11 and Rocky Linux client VMs from earlier labs for share tests.
- Your group Bookstack already contains Lab 1 and Lab 2 pages.
- TrueNAS SCALE 25.10 Community Edition ISO or the RLES template posted by the instructor.

Pre-Lab (complete before Thursday)

- Bring the snapshot + retention policy you designed in Tuesday's lecture (or draft one if you missed class).
- Confirm your Lab 1 AD domain name, a writable domain admin or join account, and the lab subnet/CIDR.
- Skim the TrueNAS 25.10 Storage and Directory Services pages linked at the end of this guide.
- Decide hostnames before you arrive. Suggested: nas.yourdomain and cronicle.yourdomain.

Recommended VM Specifications

TrueNAS SCALE (nas.yourdomain): 4 vCPU, 8 GiB RAM, 32 GiB boot disk, plus at least two 20 GiB virtual data disks. A third 20 GiB data disk is useful for the optional iSCSI extension.

Cronicle host (cronicle.yourdomain): Rocky Linux 9, 2 vCPU, 4 GiB RAM, 20 GiB disk, Internet access for Node.js 22.x packages.

Group Roles

- Member A: TrueNAS provisioning, networking, ZFS pool, and dataset layout.
- Member B: Active Directory join, SMB/NFS shares, client validation, and ACLs.
- Member C: Snapshot/replication policy, Chronicle, Zabbix/Graylog hook, and BookStack.

Every member must be able to explain the architecture, why the pool is redundant, and the difference between a snapshot and a replica. Record individual contributions on the BookStack page.

Bookstack Requirements

- Use the same shared BookStack you used in Labs 1 and 2.
- Add a Lab 3 page linked from Home. Update the architecture diagram to show TrueNAS, Cronicle, AD, clients, SMB/NFS paths, and snapshot/replication flow.
- Document while you work. Late screenshot dumps with no narrative will lose Bookstack points.

Activity Summary

- Activity 0 — Infrastructure review and appliance provisioning (15–20 min)
- Activity 1 — TrueNAS first boot, pool, and datasets (25–30 min)
- Activity 2 — Directory join and file shares (25–30 min)
- Activity 3 — Snapshots, replication, recovery, observability hook (20–25 min)
- Activity 4 — Cronicle scheduler and operational reflection (15–20 min)

Two-hour target. If an install or AD join runs long, finish the core checkpoint in class (pool ONLINE, one SMB and one NFS share working, one snapshot, Cronicle UI with one successful job) and complete polish as homework.

> **Tuesday lecture tie-in (20 minutes)**
> In pairs, design a snapshot and retention policy for a shared departmental dataset. A reasonable starting pattern is hourly for 24 hours, daily for 14 days, and weekly for 8 weeks. Decide which tasks belong in TrueNAS Periodic Snapshot / Replication versus Cronicle. Implement a simplified version of that policy in this lab.

Activity 0: Infrastructure Review & Appliance Provisioning

Purpose

Confirm Labs 1–2 are still healthy, then add a dedicated NAS and a dedicated scheduler host. Storage does not belong on identity or monitoring servers.

Procedure

1. Verify DNS, DHCP, the pfSense gateway, and that an AD user can still authenticate. Open the Zabbix and Graylog web UIs and confirm they load.
2. Provision the TrueNAS SCALE VM from the course template or the 25.10 ISO. Give it a static address or DHCP reservation that matches your Lab 1 plan. Set the hostname to nas.yourdomain (or your group equivalent).

3. Attach at least two additional 20 GiB virtual data disks. Do not put pool members on the boot disk.
4. Provision or reuse a Rocky Linux 9 VM named cronicle.yourdomain. Update packages and confirm Internet access.
5. Create DNS A and PTR records for both hosts in the zones you used in Lab 1.

Checkpoint

- TrueNAS installer or console has a stable IPv4 address and a resolvable FQDN.
- TrueNAS Storage → Disks shows the extra data disks as unused.
- cronicle.yourdomain resolves from both a Windows client and a Linux client.

Required evidence

- Figure 1 — New Hosts and Disks. TrueNAS Disks view plus hostname/date; include the unused data disks.

Activity 1: TrueNAS First Boot, Pool, and Datasets

Purpose

Finish first-boot setup, build a redundant ZFS pool, and create child datasets. Dataset purpose (Generic, SMB, Multiprotocol) changes default recordsize and ACL behavior. Choose on purpose.

Procedure

1. Complete the TrueNAS web first-boot wizard. Set a strong admin password. Point NTP at the same time source family you used in Lab 1. Set hostname and domain to match DNS.
2. Create a pool named tank (or a name that matches your domain theme). Recommended layout: two-disk mirror. If you only have two data disks, use a mirror — not a stripe. Record why you chose that vdev type.
3. Do not share the pool root. Create child datasets, for example tank/dept (SMB or Multiprotocol), tank/linux (NFS or Multiprotocol), and tank/backup as a local replication target.
4. For each dataset note share type, compression (lz4 is a sound default), and whether atime is enabled. Leave encryption off unless the instructor asks for it.
5. Confirm pool health is ONLINE and both data disks are vdev members.

Checkpoint

- Pool status is ONLINE with the expected vdev layout.
- At least two child datasets exist. None of the shares will use the pool root.
- Dashboard free space is consistent with a mirror (usable capacity is about one data disk, not two).

Required evidence

- Figure 2 — Pool Topology and Datasets. Pool status/topology and the dataset list, with hostname and date.
- Bookstack note: why a mirror versus RAIDZ1 for this lab, and why datasets were split by purpose.

Activity 2: Directory Join and File Shares

Purpose

A NAS with only local users is a workgroup box. Join TrueNAS to your Lab 1 Active Directory domain, then publish one SMB share and one NFS export. Permissions must be group-based.

Procedure

1. On TrueNAS, set DNS to your AD DNS server(s). Confirm the NAS can resolve the AD domain, a domain controller, and _ldap._tcp.yourdomain. Fix time sync before you attempt the join.
2. Directory Services → Active Directory: join the domain. Use a dedicated join account if you created one in Lab 1; otherwise use a domain admin for the lab and document that production would use a least-privilege join account. Wait until status is Healthy.
3. Create or identify two AD groups (example: nssa322-writers and nssa322-readers). Put at least one test user in writers and one user who is only a reader.
4. Create an SMB share on tank/dept. Set the dataset preset to SMB or Multiprotocol. Apply ACLs so Domain Admins (or your admin group) have Full Control, writers have Modify, and readers have Read & Execute. Enable SMB.
5. Create an NFS share on tank/linux. Restrict the export to your lab subnet. Prefer NFSv4 if both sides support it cleanly; NFSv3 is acceptable if you document the choice. Do not export to *.
6. From Windows 11, map the SMB share as the writer, create a file, then reconnect as the reader and confirm a write is denied.
7. From a Rocky client, install nfs-utils, mount the NFS export, write a test file, and add a _netdev fstab entry. If you do not have time to reboot, prove the line with mount -a after a umount.

Checkpoint

- Directory Services status is Healthy. The NAS computer object exists in AD.
- Windows writer can create a file; Windows reader cannot.
- Rocky mount type is nfs or nfs4. df -hT shows the export.

Required evidence

- Figure 3 — Directory Services Healthy. TrueNAS Directory Services status.
- Figure 4 — SMB Writer and Reader. Windows map as writer (success) and as reader (access denied), with hostname and date.
- Figure 5 — NFS Client Mount. Rocky hostname, date, mount or df -hT output, and the fstab line.

Activity 3: Snapshots, Replication, and Observability

Purpose

Mirrors protect against disk failure. They do not protect a deleted file. Snapshots give you a point-in-time restore on the same pool. Replication copies those snapshots to another dataset so a pool-level mistake is not fatal.

Procedure

1. Write a few distinctive files into the SMB and NFS datasets so snapshots have visible content.
2. Data Protection → Periodic Snapshot Tasks: create a task on tank/dept. Lab policy: every hour, retain 2 days, timestamped name. Then take one manual snapshot immediately so you have evidence during the lab window.

3. Data Protection → Replication Tasks: replicate tank/dept snapshots to tank/backup on the same NAS. Confirm at least one snapshot appears on the destination.
4. Delete or rename a test file on the live dataset. Browse or clone the snapshot and recover the file. A full dataset rollback is not required.
5. Observability hook — complete at least one: add TrueNAS to Zabbix (ICMP and/or SNMP) or produce a Graylog search that shows NAS-related or share-access events.

Checkpoint

- At least one snapshot of tank/dept is listed.
- The replication target contains a corresponding snapshot.
- You recovered the deleted or renamed file.
- TrueNAS is a Zabbix host, or you have a documented Graylog search.

Required evidence

- Figure 6 — Snapshot Task and Snapshot List.
- Figure 7 — Replication Task Status.
- Figure 8 — File Recovery. Before/after, or snapshot browse plus the restored file.
- Figure 9 — Observability Hook. Zabbix host/availability or Graylog search result.

Core required tonight vs. extension homework

Core: mirror pool, two child datasets, Healthy AD join, working SMB and NFS, one snapshot, one local replica, recovered file, Cronicle UI with one successful job, Bookstack page.

Extension: iSCSI zvol to one client; a second snapshot cadence; Cronicle calling the TrueNAS API; Zabbix widgets for pool capacity.

Activity 4: Cronicle Scheduler and Reflection

Purpose

TrueNAS already schedules snapshots well. Cronicle is for the jobs around storage: health reports, mount checks, a canary-file test, or a script that fails when a pool is not ONLINE. Your job must do real work — not only print the date.

Procedure

1. On cronicle.yourdomain install Node.js 22.x LTS. The default AppStream Node.js on Rocky 9 may be too old. Use the NodeSource 22.x repository or the method posted on myCourses. Confirm with node -v.
2. Install Cronicle as root: curl -s https://raw.githubusercontent.com/jhuckaby/Cronicle/master/bin/install.js | node — then run the setup under /opt/cronicle. Default web port is 3012.
3. Start the service, open the firewall for the web port, and keep SELinux Enforcing. Fix the port or context; do not disable SELinux.
4. Log into the Cronicle UI and change the default admin password immediately. Create one scheduled event that validates storage or a share. Acceptable jobs: mount the NFS share and test -f a canary file; smbclient a directory listing as a domain user; curl a TrueNAS API pool or alert endpoint with an API key; or logger a status line you can find in Graylog. Schedule it at least every 15 minutes, then run it once on demand.

5. As a group, walk the path: client write → dataset → snapshot → replica → Cronicle check → optional Zabbix/Graylog. Agree which tool would notice each failure first.

Checkpoint

- Cronicle UI loads and the default password has been changed.
- The event has a completed run whose log shows success, or a real failure you diagnosed and fixed.
- Every member can explain why ZFS snapshots stay in TrueNAS Periodic Snapshot Tasks rather than being driven only by Cronicle.

Required evidence

- Figure 10 — Cronicle Job. Event definition, last successful run, and a redacted snippet of the job command or script.

Required Figures (submit on Bookstack)

Use these exact titles. Every figure needs a one-line caption and must include hostname and date where a command line is shown.

- Figure 1 — New Hosts and Disks
- Figure 2 — Pool Topology and Datasets
- Figure 3 — Directory Services Healthy
- Figure 4 — SMB Writer and Reader
- Figure 5 — NFS Client Mount
- Figure 6 — Snapshot Task and Snapshot List
- Figure 7 — Replication Task Status
- Figure 8 — File Recovery
- Figure 9 — Observability Hook
- Figure 10 — Cronicle Job

Reflection Questions

Answer these on the Bookstack Lab 3 page. A few precise sentences each, with evidence.

- Q1. Why is usable capacity on your two-disk pool closer to one disk than two? What failure does that layout protect, and what failure does it not protect?
- Q2. Why did you create child datasets instead of sharing the pool root?
- Q3. What broke, or would have broken, if DNS or time had been wrong during the AD join?
- Q4. In one paragraph: snapshots are not backups. How does your replication task change that statement, and what would still be missing for a production 3-2-1 design?
- Q5. Which jobs belong in TrueNAS Periodic Snapshot / Replication, and which belong in Cronicle? Use your Tuesday policy and what you actually implemented.

What You Submit

- Bookstack Lab 3 page URL (put it on the cover fields at the top of this guide if your instructor collects a paper/PDF copy).

- Updated architecture diagram.
- Figures 1–10 with captions.
- Answers to reflection questions Q1–Q5.
- Troubleshooting journal: at least two detailed entries (problem, what you tried, fix).
- Individual contribution notes for each member.
- Links to the official TrueNAS 25.10 and Cronicle documentation you actually used.

You do not submit this Word file unless the instructor asks for a printed cover sheet. Bookstack is the official deliverable.

Conclusion

You now have shared, recoverable storage and a scheduler on top of Labs 1 and 2. You practiced the decisions that matter in production: isolate the NAS, do not share the pool root, join the directory for SMB, restrict NFS by network, snapshot for operational recovery, replicate so snapshots are not the only copy, and automate the checks humans forget to run.

Leave TrueNAS and Cronicle running. Later labs will expect a place to put web content, mail data, or container volumes — and they will expect those datasets to be monitored.

Rubric

Total 100 points. Passing threshold is 70.

Item	Points	Sign-off
TrueNAS deployed: resolvable hostname, pool ONLINE, datasets not on pool root (Fig. 1–2)	**15**	
AD join healthy + SMB share with group ACL; writer vs reader shown (Fig. 3–4)	**20**	
NFS export mounted from Rocky Linux; fstab/_netdev documented (Fig. 5)	**10**	
Periodic snapshot + at least one snapshot + file recovery (Fig. 6, 8)	**15**	
Local replication task with snapshot on the target dataset (Fig. 7)	**10**	
Cronicle deployed; one operational job ran successfully (Fig. 10)	**15**	
Bookstack: diagram, Q1–Q5, journal, contributions, Fig. 9 hook	**15**	
Total	**100**	

Common Pitfalls & Troubleshooting

- AD join fails — almost always DNS or time. Confirm AD DNS, _ldap._tcp.yourdomain, and NTP within a few seconds of the DC.
- SMB works for a local user but not a domain user — Directory Services is not Healthy, or the dataset ACL never received the domain group.
- Cannot write to SMB — root/admin is not a valid SMB user. Use a domain account and a group ACE.
- NFS mount hangs — wrong subnet on the export, pfSense filtering 2049/tcp (and rpcbind/mountd for NFSv3), or a hostname that does not match the export.

- Pool looks “half empty” — a two-disk mirror has about one disk of usable space. That is redundancy working.
- Snapshots exist but replication is empty — dataset paths do not match, or the task never ran after the first snapshot.
- Cronicle installer fails — Node.js is too old. Check node -v is 22.x or later.
- Cronicle UI unreachable — firewalld, listen address in /opt/cronicle/conf, or the service is down. Use ss -lntp | grep 3012.
- Sharing the pool root — creates painful permission and snapshot problems later. Always share a child dataset.

Optional Extension (homework)

- Present a zvol over iSCSI to one client and compare it with SMB/NFS (who owns the filesystem?).
- Create a TrueNAS API key and have Cronicle call /api/v2.0/pool or /alert/list, failing the job if the pool is not ONLINE.
- Add Zabbix widgets for NAS availability and pool capacity next to your Lab 2 dashboard.

Primary Resources

TrueNAS 25.10 Documentation Hub: https://www.truenas.com/docs/scale/25.10

TrueNAS SMB shares: https://www.truenas.com/docs/scale/25.10/scaletutorials/shares/smb/

Cronicle setup: https://github.com/jhuckaby/Cronicle

Cronicle install (Node 22+ required): https://github.com/jhuckaby/Cronicle/blob/master/docs/Setup.md

Excellent work — your infrastructure can now store, share, protect, and check itself.